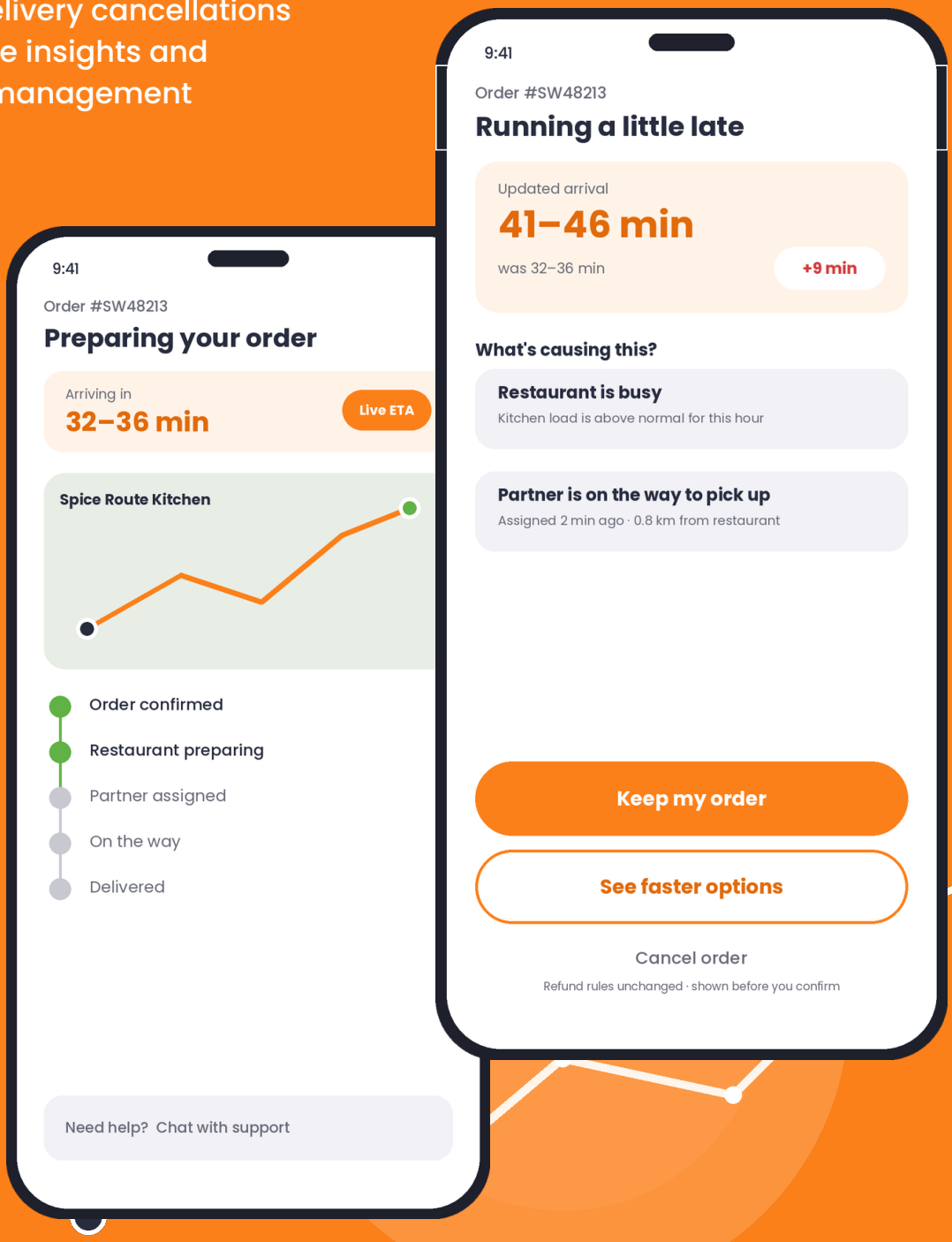
PRODUCT MANAGEMENT CASE STUDY

Swiggy SmartSave

Reducing food delivery cancellations through predictive insights and intelligent order management



A product strategy and user experience case study
Independent portfolio project, not affiliated with Swiggy.
All figures are proposals or illustrative scenarios unless stated.

EXECUTIVE SUMMARY

Prevent the cancellation, not just the refund

Most cancellation work in food delivery starts after the customer taps cancel. SmartSave starts earlier. It predicts which orders are drifting toward failure, tells the customer the truth while there is still time to act, and only then offers choices.

−20%
target reduction in order cancellation rate (pilot hypothesis)

1,600
orders per month potentially saved in an illustrative 100,000-order pilot

3
candidate solutions scored with RICE; Smart ETA selected as the MVP

The recommendation

Build **Phase 1: Predictive ETA and proactive delay alerts** first. It scores highest in our illustrative RICE model, reaches the broadest set of orders, fits inside the current tracking flow and creates the data foundation for Phase 2 (alternatives) and Phase 3 (restaurant and partner coordination).

What this report is, and is not

Data integrity note. No survey percentages appear in this report because no survey has been run yet. Baselines, targets, RICE inputs and the business-impact scenario are labelled as proposals or hypotheses. The customer survey in Section 4 is designed to be run first; charts and conclusions are added only from real responses.

Contents

01	Overview and why it matters
02	Problem definition and research grounding
03	Goals, non-goals and success metrics
04	Problem validation and survey plan
05	User personas
06	Proposed solutions and trade-offs
07	Product design: journey, wireframes, alert logic
08	Prioritisation with RICE and the roadmap
09	Risks and experiment design
10	Business impact and conclusion
A–C	Questionnaire, assumptions register, references

SECTION 01

Overview and why it matters

A cancelled order is a failed promise made visible. It has four victims.

Company: Swiggy. **Product:** the Swiggy food delivery app. **Proposed feature:** SmartSave, an intelligent order-management layer that identifies cancellation risk early and helps customers, restaurants and delivery partners resolve problems before an order fails.

Objective: reduce avoidable cancellations by finding their causes, improving delivery transparency, predicting order failures and giving customers proactive, honest options.

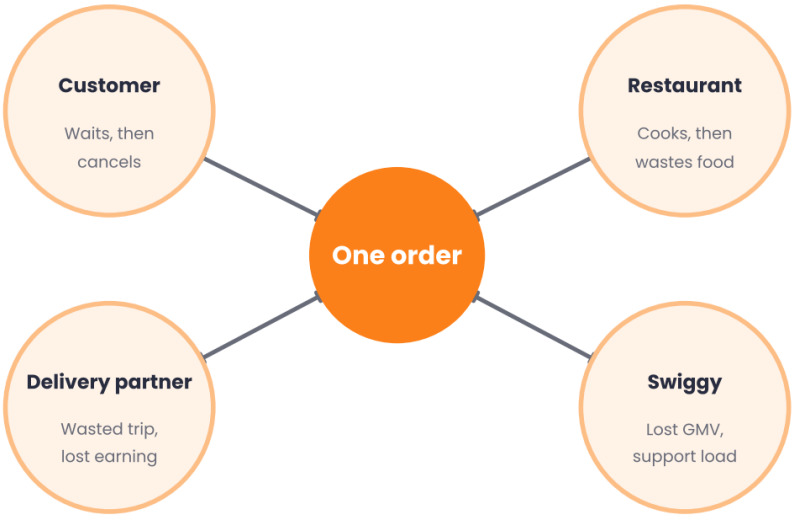


Figure 1. One cancelled order creates cost for four parties at once.

The scenario

A customer orders lunch. The restaurant accepts, but preparation runs long. A delivery partner is assigned late. The ETA keeps creeping up. The customer, now unsure whether food will arrive at all, cancels. The restaurant has already cooked the meal, the partner may have already travelled, and the platform has lost an order and some trust.

Stakeholder	What a cancellation costs them
Customer	Time wasted, uncertainty about food and refund, erosion of trust
Restaurant	Lost revenue and, if food is already prepared, wastage
Delivery partner	Wasted trip, idle time, lost earning opportunity
Swiggy	Lost order value, support contacts, weaker repeat ordering and operational inefficiency

The central product opportunity is to **prevent avoidable cancellations** rather than process them better after they happen.

SECTION 02

Problem definition and research grounding

Cancellations are not one problem. They are several problems that happen at different moments.

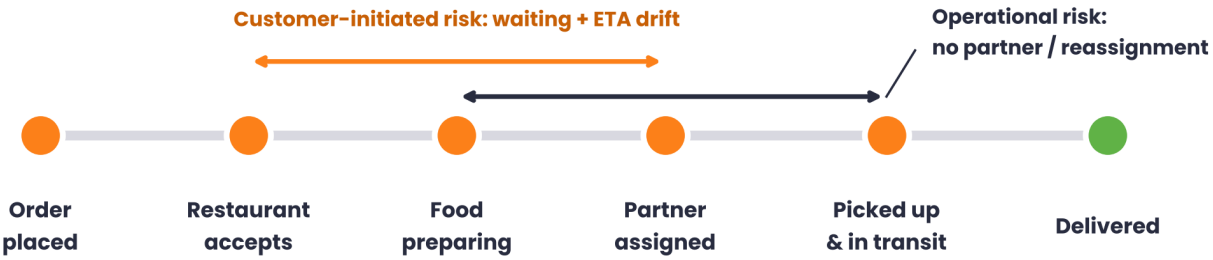


Figure 2. Order lifecycle with the two cancellation risk windows we will analyse separately (working hypothesis).

Cancellation stage	Possible reason
Before restaurant acceptance	Restaurant unavailable or rejects the order
During food preparation	Long preparation time or ingredient shortage
During delivery assignment	No delivery partner available
After delivery assignment	Partner unavailable or reassigned
In transit	Excessive delay or incorrect location
Customer-initiated	Long wait, changed plans or wrong order

The stage taxonomy is a set of proposed categories for research. It is not verified Swiggy cancellation data.

What published research already tells us

Two peer-reviewed studies on on-demand food platforms shape our hypotheses. Liang et al. (2025, *Transportation Research Part A*) analyse cancellations across two waiting stages, waiting for matching and waiting for pick-up. They find that accumulated waiting time affects cancellation differently before and after the expected waiting time, which points to customers judging a wait against the expectation they were given. They also find that a familiar driver lowers cancellation risk.

Xu, Yan and Tong (2021, *Journal of Business Research*) find that longer waiting during service production, rather than during delivery, increases cancellations on a food platform. Together the papers suggest that **expectations set by the ETA** and **delays before pickup** deserve first attention. That is exactly where Smart ETA operates.

Hypotheses to validate

- Customers cancel more when the estimated delivery time keeps increasing.
- Restaurant preparation delays are a leading source of dissatisfaction.
- Late partner assignment raises cancellation likelihood.
- Missing, vague or stale order updates reduce customer confidence.
- Proactive communication and feasible alternatives can prevent some avoidable cancellations.

SECTION 03

Goals, non-goals and success metrics

Being explicit about what we will not do is as important as what we will.

Goals

- Reduce avoidable customer-initiated and operational cancellations
- Improve delivery-time accuracy and transparency
- Identify high-risk orders before cancellation
- Improve restaurant and partner coordination
- Increase customer trust and repeat ordering

Non-goals

- Eliminate every cancellation
- Force customers to accept delayed orders
- Penalise partners for delays outside their control
- Replace refund and support systems
- Guarantee a fixed delivery time regardless of conditions

Success metrics

Proposed North Star: completed orders per 100 orders placed, within the promised ETA window. It rewards fewer cancellations and honest ETAs together, so the team cannot improve one by degrading the other.

Metric	Baseline	Proposed target	Type
Order cancellation rate	To be measured	-20%	Primary
Customer-initiated cancellations	To be measured	-25%	Primary
Late deliveries	To be measured	-15%	Secondary
Order-tracking engagement	To be measured	+20%	Secondary
Cancellation-related support contacts	To be measured	-20%	Secondary
Partner waiting time, restaurant rejections	To be measured	No worse than control	Guardrail

Targets are starting hypotheses for the case study, not Swiggy performance figures. The baseline is set from order-level data before any improvement is claimed.

SECTION 04

Problem validation and survey plan

Survey for stated preference, interview for the why, order logs for the real baseline.

50–100
survey respondents

8–10
stakeholder interviews

3
groups: customers, restaurants, partners

5
hypotheses to prove or kill

Hypothesis	Method	Survey question
Delivery delays are a major cancellation trigger	Survey and order-log analysis	Q4, Q5
Customers want accurate ETAs at order time	Survey and interviews	Q7
Customers prefer alternatives to cancelling	Survey, then A/B test	Q8
Proactive updates can prevent cancellations	Survey, then A/B test	Q9
Tracking improvements are valuable	Survey and usability test	Q10

Analysis plan: the charts we will build

Chart	Question	Decision it informs
Reasons for cancellation (pie)	Q4	Is delay the leading cause?
Cancellation frequency (bar)	Q3	How common is the behaviour?
Preferred SmartSave features (horizontal bar)	Q10	Feature order for roadmap
Willingness to wait after a delay (stacked bar)	Q6, Q9	Alert thresholds and copy
Satisfaction with existing tracking (bar)	Q7	Size of the transparency gap

Decision rules (set before looking at data)

- Smart ETA remains the MVP if delay-related reasons rank in the top two causes (Q4, Q5) **and** at least half of respondents say an updated ETA would make them wait (Q9).
- If alternatives clearly beat updated ETAs in Q8, bring Solution 2 forward.
- Report sample size, sampling method and likely bias with every chart.

A survey measures *reported* experience and preference. It does not establish the platform-wide cancellation rate. That figure comes only from order-level logs.

SECTION 05

User personas

Three illustrative people who experience the same failure from different seats.



Rahul, 25

Software engineer, the busy professional

PAIN POINT

Orders lunch at his desk. Repeated ETA changes disrupt his workday.

IDEAL SOLUTION

Accurate delivery estimates and an early delay alert.



Priya, 21

College student, the budget-conscious

PAIN POINT

Orders discounted meals and dislikes uncertainty after paying, especially about refunds.

IDEAL SOLUTION

Alternative suggestions and transparent refund updates.



Amit, 34

Restaurant owner, the partner

PAIN POINT

Rush-hour pressure. Sometimes cooks before a delivery partner is assigned.

IDEAL SOLUTION

Realistic preparation-time management and earlier partner dispatch.

Personas are illustrative and guide the research plan. They are not interviewed individuals.

Key takeaways

- Customers need reliable delivery information.
- Restaurant partners need better order coordination.
- Delivery partners need efficient assignment and pickup management.
- Preventing cancellations takes collaboration across the whole ecosystem.

SECTION 06

Proposed solutions and trade-offs

Three ways to attack the problem: inform, offer choice, or fix operations.

SOLUTION 1

Smart ETA prediction and delay alerts

Use historical data and real-time operational signals to estimate more realistic delivery times and warn the customer early.

How it works

1. Estimate food preparation time
2. Consider delivery-partner availability
3. Incorporate pickup and travel conditions
4. Notify the customer when a meaningful delay is predicted
5. Provide a revised delivery estimate

Advantages

- Better transparency
- Lower uncertainty
- Earlier customer intervention

Disadvantages

- Needs reliable real-time data
- Predictions can still be wrong

SOLUTION 2

Smart alternative recommendation

When an order faces a major disruption, recommend suitable alternatives and let the customer decide.

How it works

1. Identify unavailable items or severe delays
2. Find similar restaurants or dishes
3. Show alternative ETAs and prices
4. Let the customer switch or keep waiting
5. Explain refund or payment changes before confirmation

Advantages

- Preserves purchase intent
- Gives customers control
- Can recover some lost orders

Disadvantages

- Alternatives may not exist
- Switching complicates payment and fulfilment

SOLUTION 3

Restaurant and delivery-partner coordination

Align food-ready time with partner arrival and escalate high-risk orders to operations.

How it works

1. Predict restaurant preparation time
2. Estimate when a partner should reach the restaurant
3. Match partners by proximity and availability
4. Flag orders with assignment or preparation delays
5. Trigger operational intervention for high-risk orders

Advantages

- Reduces coordination gaps
- Improves pickup efficiency
- Cuts operational cancellations

Disadvantages

- Requires restaurant-side adoption
- Complex operational integration

Trade-off analysis

Criteria	1 · Smart ETA	2 · Alternatives	3 · Coordination
Customer impact	High	High	Medium
Implementation complexity	Medium	Medium	High
Operational impact	Medium	Low	High
Data dependency	High	Medium	High
Scalability	High	High	Medium

Qualitative PM assessments, not measured results. Green is favourable and red unfavourable for the criterion.

SECTION 07

Product design: journey, wireframes and alert logic

SmartSave is a thin layer on the existing tracking experience, not a new app.

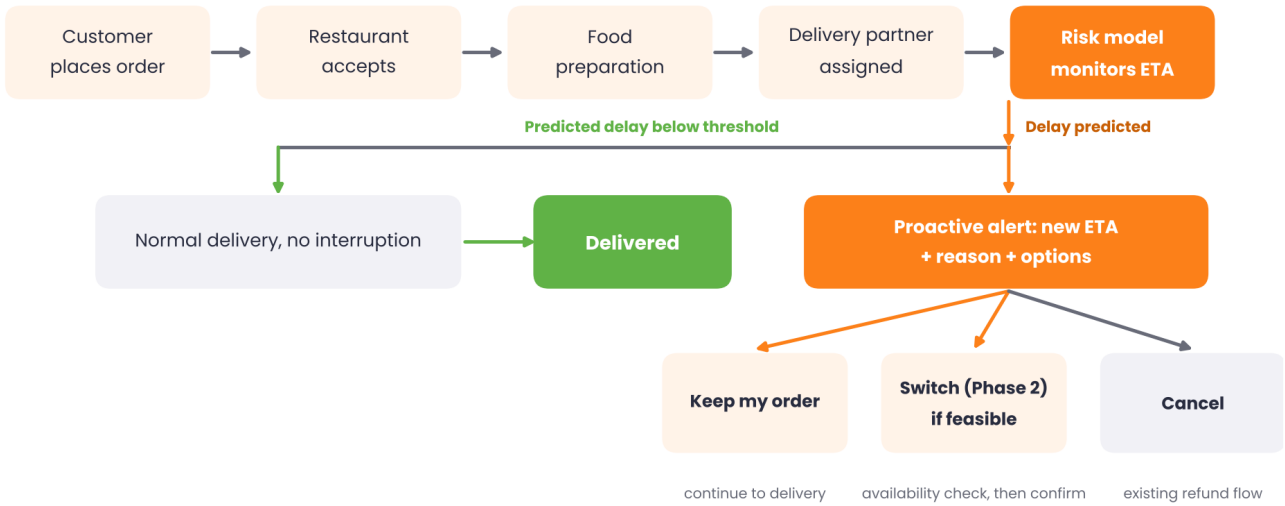


Figure 3. Proposed user journey. Normal orders see no change; only predicted delays trigger the alert.

Design principles

- **Lead with the new ETA and the reason.** Customers forgive delay faster than silence.
- **Make "Keep my order" the primary action.** Cancel stays visible, honest and unchanged.
- **Show refund impact before any switch is confirmed.** No silent payment changes.
- **Alternatives appear only when genuinely feasible.** A bad suggestion is worse than none.

Wireframes

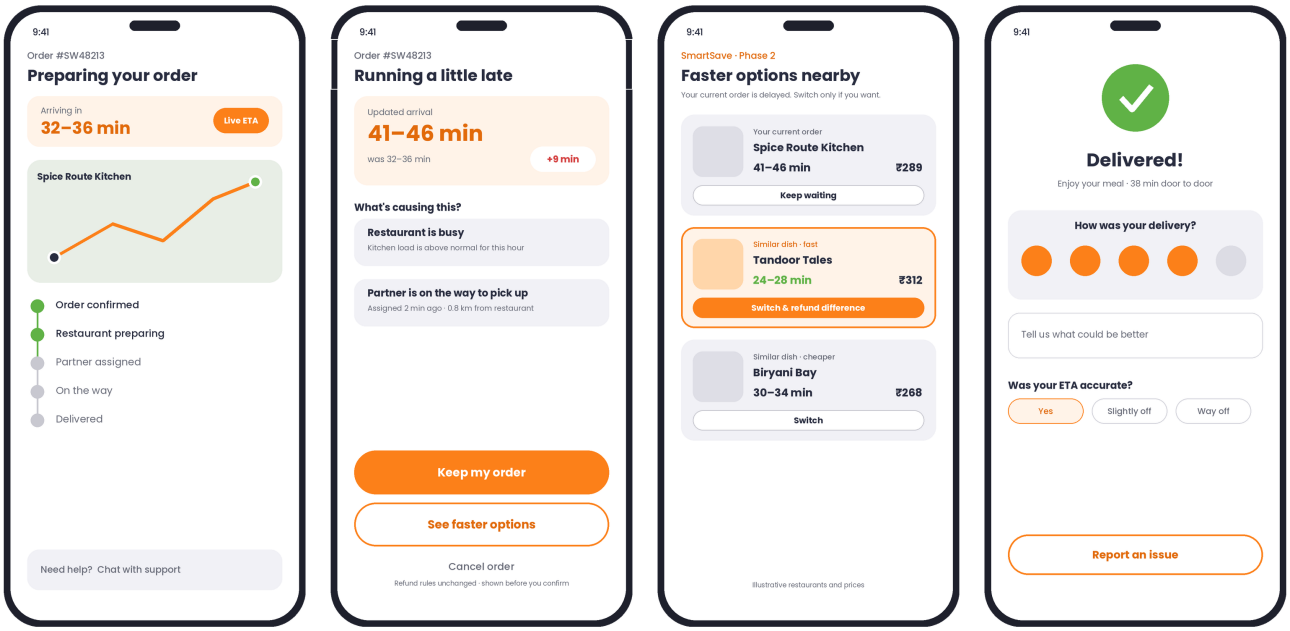


Figure 4. Illustrative wireframes (not Swiggy screenshots). Restaurant names and prices are placeholders.

User story and acceptance criteria

> As a Swiggy customer, I want an early notification when my food delivery is likely to be delayed, so that I can make an informed decision rather than cancelling out of uncertainty.

- A notification is sent when a meaningful delay is predicted.
- The updated ETA is displayed with the reason where known.
- The customer can continue waiting with one tap.
- Alternative options appear only when genuinely feasible.
- Existing cancellation and refund rules remain transparent.

Proposed alert logic (Phase 1)

Illustrative design for the prediction and notification layer, to be refined with data science and engineering. Nothing here describes Swiggy's current systems.

Component	Proposal
Inputs	Restaurant preparation history and current kitchen load, partner availability and distance, travel and traffic conditions, time of day, order size and complexity
Output	A predicted delivery window with a confidence level, refreshed as the order progresses
Trigger	Notify only when the predicted delay exceeds a threshold **and** confidence is sufficient. Thresholds are tuned in experiment, not fixed in advance
Frequency cap	At most one proactive alert per order plus one resolution message, to avoid notification fatigue
Calibration	Track predicted versus actual delivery time continuously. Prefer honest ranges to falsely precise points
Fallback	If signals are unreliable, show the existing ETA without alerts

SECTION 08

Prioritisation with RICE and the roadmap

A framework to disagree with data instead of opinions.

RICE = (Reach × Impact × Confidence) ÷ Effort. The inputs below are hypothetical product-team estimates for demonstration, not Swiggy data.

Factor	Smart ETA	Alternatives	Coordination
Reach (monthly units)	10,000	7,000	8,000
Impact (1–3)	2	3	3
Confidence	0.8	0.7	0.6
Effort (person-months)	4	5	8
RICE score	**4,000**	**2,940**	**1,800**

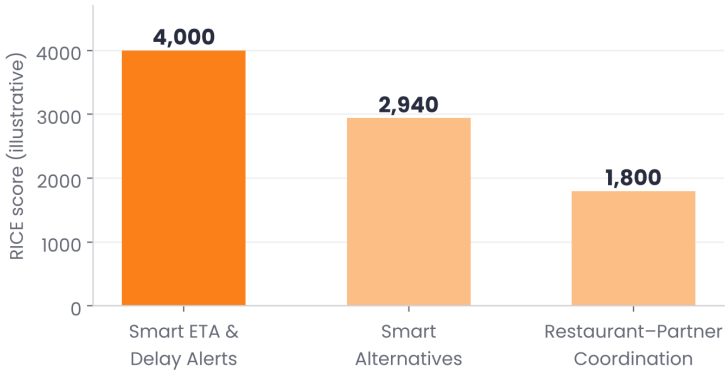


Figure 5. Illustrative RICE scores. Orange marks the selected solution.

Why Smart ETA first

- It directly addresses uncertainty during delivery, the moment customers decide to cancel.
- It reaches the widest range of orders.
- It fits into the existing flow without changing how people order.
- It produces the prediction signals that Solutions 2 and 3 depend on.
- It works alongside current restaurant and partner operations.

Phased roadmap



Figure 6. Phase 1 proves the core belief; later phases reuse its data and learning.

Launching all three together would multiply integration risk and blur what actually moved the metric. Sequencing keeps the experiment clean.

SECTION 09

Risks and experiment design

Assume the feature can make things worse, then design the test to catch it.

Risk	Potential impact	Mitigation
Incorrect ETA predictions	Loss of customer trust	Continuously recalibrate; show ranges rather than false precision
Excessive notifications	Notification fatigue	Alert only on meaningful delay; cap alerts per order
Restaurant data inaccuracies	Wrong preparation estimates	Improve restaurant-side status updates
Delivery partner unavailability	Continued delays	Improve assignment and reassignment logic
Low feature adoption	Limited impact	Contextual prompts and simple UX
Operational complexity	Higher implementation cost	Launch an MVP in selected cities

A/B test design

Control (A)
Existing order-tracking experience.

Treatment (B)
Predictive ETA, delay notification and the option to keep waiting.

Metric class	Metrics
Primary	Cancellation rate among eligible orders
Secondary	On-time delivery rate; customer satisfaction; refund and support contact rate; notification engagement; repeat-order rate
Guardrail	Delivery-partner waiting time; restaurant order rejection rate; average delivery time; customer complaints

Sample size, worked example

Assume an 8% baseline cancellation rate among eligible orders and a 20% relative reduction (8% to 6.4%). For a two-sided test at 5% significance and 80% power, a standard two-proportion calculation gives roughly **4,100 eligible orders per arm**. If the real baseline differs, the number must be recomputed. Eligible orders are those where the model predicts a meaningful delay.

Experiment hygiene

- Randomise at order or customer level where operationally feasible; compare equivalent cohorts and cities.
- Analyse customer-initiated cancellations separately from restaurant and platform cancellations.
- Pre-register the primary metric and decision rule; check guardrails before celebrating.
- Roll out in stages (small share of eligible orders, then wider) only after guardrails hold.

SECTION 10

Business impact and conclusion

What we expect, what it is worth, and why this is the right first bet.

Illustrative impact scenario

Assume a pilot processes 100,000 orders a month at a hypothetical 8% cancellation rate. A 20% relative reduction gives: 100,000 × 8% = 8,000 cancellations; 8,000 × 20% = **1,600 potentially prevented cancellations per month**.

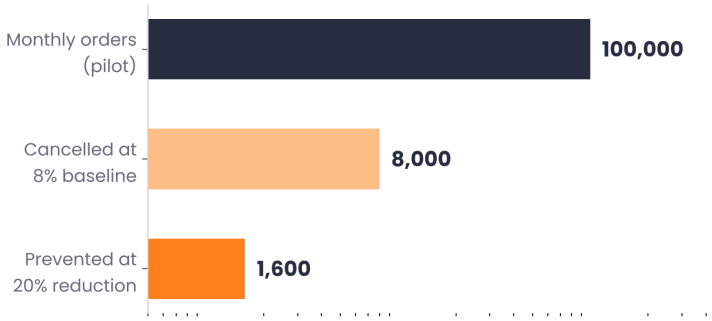


Figure 7. Illustrative monthly impact (log scale). A scenario, not a Swiggy forecast.

Business metric	Proposed pilot improvement
Order cancellation rate	20% reduction
Late delivery rate	15% reduction
Cancellation-related support contacts	20% reduction
Customer satisfaction	15% improvement
Repeat-order rate	10% improvement

Pilot targets to be validated through experimentation.

Expected benefits

Customers
- More accurate ETAs
- Less uncertainty
- Control over delayed orders

Restaurants
- Better order coordination
- Fewer avoidable cancellations
- Less food wastage

Delivery partners
- Better assignment coordination
- Fewer wasted pickup trips
- Better planning

Swiggy
- Customer retention
- Fulfilment efficiency
- Higher completed-order volume

Conclusion

SmartSave applies one idea: tell customers the truth early, then give them real choices. Instead of waiting for frustration to turn into a cancellation, it detects the delay, explains it and keeps the customer in control. The recommended path is to ship Smart ETA and delay alerts as a contained MVP in selected cities, prove the effect with a properly powered A/B test, and let the evidence decide whether alternatives and coordination come next.

Immediate next steps: (1) run the 10-question survey and 8–10 interviews; (2) pull order-level data to set real baselines by cancellation stage; (3) review RICE inputs with data science and operations; (4) prototype the delay screen and usability-test it; (5) finalise experiment design and guardrails.

APPENDIX A

Survey questionnaire (Google Forms)

Keep the form under three minutes. Add a screener for recent online ordering, and capture city and age band for cuts.

#	Question	Answer options	Tests
Q1	How often do you order food online?	Daily, a few times a week, weekly, monthly, rarely	
Q2	Which food delivery apps do you use most?	Multi-select	
Q3	How many orders did you cancel in the last 3 months?	None, 1–2, 3–5, 6 or more	Frequency
Q4	What was the main reason for your most recent cancellation?	Delay, ETA kept increasing, restaurant delay, no partner assigned, ordered by mistake, changed plans, other	H1
Q5	At which stage did you cancel?	Right after ordering, during preparation, while waiting for a partner, after pickup or in transit	H1
Q6	How long past the first ETA before you consider cancelling?	Under 10 min, 10–20, 20–30, over 30	Threshold
Q7	How important is an accurate ETA when you order?	Scale 1–5	H2
Q8	If your order is delayed, what would you prefer?	Keep waiting with updated ETA, switch to an alternative, cancel with instant refund, contact support	H3
Q9	Would an early alert with the reason make you wait longer?	Definitely yes to definitely not	H4
Q10	Which improvement would help you most?	More accurate ETA, live partner tracking, delay reason, alternatives, refund transparency	H5

APPENDIX B

Assumptions register

Item	Status	How it gets resolved
Baseline cancellation rate by stage	Unknown (8% used only in the scenario)	Order-level data pull
Targets (-20%, -25%, -15% and others)	Hypothesis	A/B test readout
RICE reach, impact, confidence, effort	Hypothetical estimates	Review with data science, ops and engineering
Survey results	Not yet collected	Run Google Form, then add real charts
Stage taxonomy of cancellations	Working hypothesis	Validate against logs and interviews
Alert thresholds and frequency cap	Proposal	Tune in experiment

APPENDIX C

References

- Liang, J., Zhao, Y., Wang, H., Yang, L. and Ke, J. (2025). Understanding order cancellation behavior in on-demand delivery services. *Transportation Research Part A: Policy and Practice*, 198. doi:10.1016/j.tra.2025.104515
- Xu, X., Yan, N. and Tong, T. (2021). Longer waiting, more cancellation? Empirical evidence from an on-demand service platform. *Journal of Business Research*, 126, 162–169.
- Swiggy Terms and Conditions and Customer Care pages (swiggy.com): official information on fulfilment, approximate delivery estimates and support.
- Swiggy Annual Report 2025 and Swiggy press release "How India Swiggy'd 2025": business context and ordering trends. Review the originals before quoting figures.

Tools suggested for execution: Google Forms (survey), Excel (analysis and KPI maths), Power BI (cancellation dashboard), Figma (high-fidelity wireframes), PowerPoint (final presentation), Python optional (exploratory analysis and ETA modelling). Frameworks used: RICE, personas, journey mapping, MVP prioritisation, A/B testing.